

AI Intelligence Platform for Data Centre EPC Project Delivery

Professional Project Report

Abstract

An AI-powered EPC intelligence platform that integrates project documents, procurement, schedules, RFIs, quality records, and commissioning workflows into a centralized intelligent system.

Introduction

Modern EPC projects generate massive volumes of technical data. This platform applies AI, RAG, predictive analytics, and computer vision to automate compliance, detect risks, and improve project delivery.

Problem Statement

Information fragmentation, procurement delays, manual document reviews, commissioning failures, and schedule overruns affect project success.

Objectives

Automate quality compliance, predict schedule risks, improve supply-chain visibility, assist commissioning engineers, and provide AI-powered document search.

Modules

1. Specification & Quality Compliance Agent
2. Predictive Schedule Risk Engine
3. Supply Chain Visibility Agent
4. Commissioning Quality Assurance Copilot
5. Project Knowledge & RFI Intelligence Agent

System Architecture

Users → Web Portal → Backend API → AI Engine → ML Models → Vector Database → MongoDB/PostgreSQL → Dashboard & Reports

Technology Stack

Frontend: React.js

Backend: Node.js & Express.js

Database: MongoDB/PostgreSQL

AI: Python, TensorFlow, Scikit-learn, LangChain, FAISS/ChromaDB, OpenAI-compatible LLMs

Database

Users, Projects, Documents, RFIs, Suppliers, Risks, Test Records, Quality Logs.

Advantages

Reduced manual effort, faster RFI handling, predictive risk alerts, automated compliance, improved commissioning quality, centralized knowledge.

Future Scope

BIM integration, IoT monitoring, Digital Twin, Drone inspection, AR/VR visualization, Mobile application.

Conclusion

The platform enables intelligent, scalable, and efficient EPC project delivery by combining AI with construction project management.